NUI Galway
OÉ Gaillimh

Autumn Examinations 2011/2012

Exam Codes	3BS4, 4BME1, 4FM2
Exam	Third Science, Fourth Mathematics and Education, Fourth Financial Mathematics and Economics
Module	Nonlinear Systems
Module Code	MP491
Paper No.	
Repeat paper	Yes
External Examiner	Prof. Giuseppe Saccomandi
Internal Examiner	Prof. Michel Destrade
Instructions	Full marks for THREE completed questions [**20 marks each**]
Duration	TWO HOURS
No. of Pages	3 pages
Discipline	Applied Mathematics
Requirements	Release to Library: No Statistical Tables: Yes - Log Tables Other material: Yes - Calculator

1. (a) Consider the one-parameter family of non-linear ordinary differential equations

$$\frac{dx}{dt} = (x^2 - 1)(x^2 + \mu),$$

where μ is a real parameter.

i. Investigate the stability behaviour of all equilibrium points. [**8 marks**]

ii. Sketch the bifurcation diagram, and indicate the behaviour of the non-equilibrium solutions. [**3 marks**]

iii. Find all bifurcation points and classify them. [**2 marks**]

(b) Consider the non-linear system of ordinary differential equations

$$\begin{aligned} \dot{x} &= 2\mu x + 2y + 2xy^2, \\ \dot{y} &= -2x + 2\mu y + 2y^3. \end{aligned}$$

i. Find the value of μ for which a Hopf bifurcation occurs at the origin $(x, y) = (0, 0)$. How does the nature of the equilibrium point change at the Hopf bifurcation? [**5 marks**]

ii. Show that $(0, 0)$ is the only equilibrium point. [**2 marks**]

2. Consider the linear system of ordinary differential equations

$$\frac{dx}{dt} = 2x - 4y,$$

$$\frac{dy}{dt} = x + \mu y,$$

where $x = x(t)$, $y = y(t)$ and μ is a real parameter.

(a) Describe how the nature of the equilibrium point $(0, 0)$ changes as $\mu \in \mathbb{R}$ varies. In particular, check that $(0, 0)$ cannot be an attractor equilibrium point for any value of μ. [**14 marks**]

(b) Do a rough sketch of the phase-plane portrait about the equilibrium point $(0, 0)$ in the case $\mu = 4$. Using arrows, indicate in which way the trajectories are described as t increases. [**6 marks**]

3. Consider the second order nonlinear ordinary differential equation

$$\ddot{x} - x^2 - x = 0.$$

(a) Show that the second order nonlinear ordinary differential equation can be written as **[2 marks]**

$$\begin{aligned} \dot{x} &= y, \\ \dot{y} &= x^2 + x. \end{aligned}$$

(b) Show that this system is Hamiltonian and find the Hamiltonian function $H(x, y)$. **[4 marks]**

(c) Find all equilibrium points and classify them. **[4 marks]**

(d) Use the Hamiltonian function to sketch the phase-plane portrait, as follows.

i. Sketch some trajectories in the neighborhood of the equilibrium points; **[2 marks]**

ii. Study the level curves of H passing through the equilibrium points: determine where they are defined, where they cross the coordinates axes and in which direction they are described; **[6 marks]**

iii. Sketch those trajectories and a few more to illustrate the phase-plane portrait for this system. **[2 marks]**

4. Consider the non-linear system of ordinary differential equations

$$\begin{aligned} \dot{x} &= xy + x + y + 1, \\ \dot{y} &= xy - x - y + 1. \end{aligned}$$

(a) Locate and classify all equilibrium points in the associated phase plane. **[4 marks]**

(b) Identify, and sketch separately, the isoclines corresponding to $k = -1, 0, 1, \infty$. **[12 marks]**

(c) By using all four isoclines in (b), sketch a sufficient number of phase plane curves to illustrate the phase-plane portrait for this system. **[4 marks]**